

BAO-NGOC PHAN

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EDUCATION

Le Hong Phong High School for the Gifted

Ninh Binh, Vietnam

Specialization: Chemistry

2023 – 2026

GPA: 9.5/10.0 · AP Chemistry 5 · AP Calculus AB & BC 5

- Achieved the only perfect 10 in specialized Chemistry in the 30-year history of the High School Entrance Exam.
- Received certificates of Merit from both the Nam Dinh City Youth Union and Nam Dinh Provincial Department of Education and Training for outstanding academic performance and active participation in Youth Union activities.

RESEARCH EXPERIENCE

Center for Computational Science @HNUE

Hanoi, Vietnam

Research Assistant

Sep 2025 – present

Project: Tuning the Emission and Circularly Luminescence Dissymmetry of CzBN-Based Heterohelicenes

(under review – 3rd author)

Supervisor: Professor Hue Minh Thi Nguyen

- Performed density functional theory (DFT) calculations using Gaussian and VASP for research projects; prepare figures, tables, and visualizations for reports, academic papers, and presentations; perform statistical analyses and interpret results to support research hypotheses.

Lumiere Research

Mar 2026 – May 2026

Project: Evaluating surface functionalization strategies in carbon nanotube-based drug delivery for cancer therapy: A review

Advisor: Dr. Simran Kerai

- 100% financial aid; conducted a systematic literature review on carbon nanomaterials in drug delivery, biosensing, and biomedical diagnostics; analyzed structure–property relationships and functionalization strategies to improve nanomaterial biocompatibility.

SKILLS

Technical Skills: Computational Chemistry (Gaussian 09, VASP (beginner), Chemcraft); Programming (JavaScript, Python, C++); Development Tools (Git, GitHub, VS Code); Graphic Design (Figma).

Other Skills: Academic tutoring, Piano performance (intermediate level, 10 years of experience), Photography, Writing (creative)

Language: Vietnamese (native); English (IELTS 7.0); Mandarin (TOCFL A1).

TEACHING ASSISTANT

HChemO

Hanoi, Vietnam

Teaching Assistant

July 2025 – present

- Developing supplemental study materials and practice problems to reinforce learning; supporting teachers in attendance tracking, grading, and answering students' questions; providing targeted tutoring on AP Chemistry topics, including kinetics, equilibrium, and electrochemistry.

LEADERSHIP & ACTIVITIES

- Founded ChemOlyms, a student-led chemistry mentorship focused on Olympiad training in Vietnam. *April 2025 – July 2026*
- Founded Louisa's Artistry, a page sharing photography knowledge and photos. *November 2024 – present*
- Lead Pianist, School Band - responsible for arranging and performing in live shows. *August 2023 – July 2026*
- Designed and taught chemistry for underprivileged children (Chemistry Instructor, SOS Children's Village). *June 2023 – August 2024*

ADDITIONAL AWARDS/HONORS

- Gold Medal (Rank 3rd), Asian Chemistry Olympiad (2025)
- Silver Medal, Chemico (Stemco by NUS) (2025)
- Bronze Medal, Canadian Chemistry Contest (CCC) (2025)
- Merit Award, Vietnam National Science – Chemistry Olympiad (2024)
- Gold Medal, C3L6 by University of Cambridge (2025)
- Second Prize, Provincial Excellent Student Competition in Chemistry (Grade 12, taken as an 11th grader, 2024)
- High Distinction, International Chemistry Quiz (ICQ) (2025)
- LHP Talent Scholarship (2025)
- Bronze Medal, Hanoi University of Science Olympiad (HSGSO) (2024)
- First Prize, Le Hong Phong Chemistry Research Conference (2025)